Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

2026-01-09

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0.1 Summary

- Humans produce higher novelty (surprisal) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- AI contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

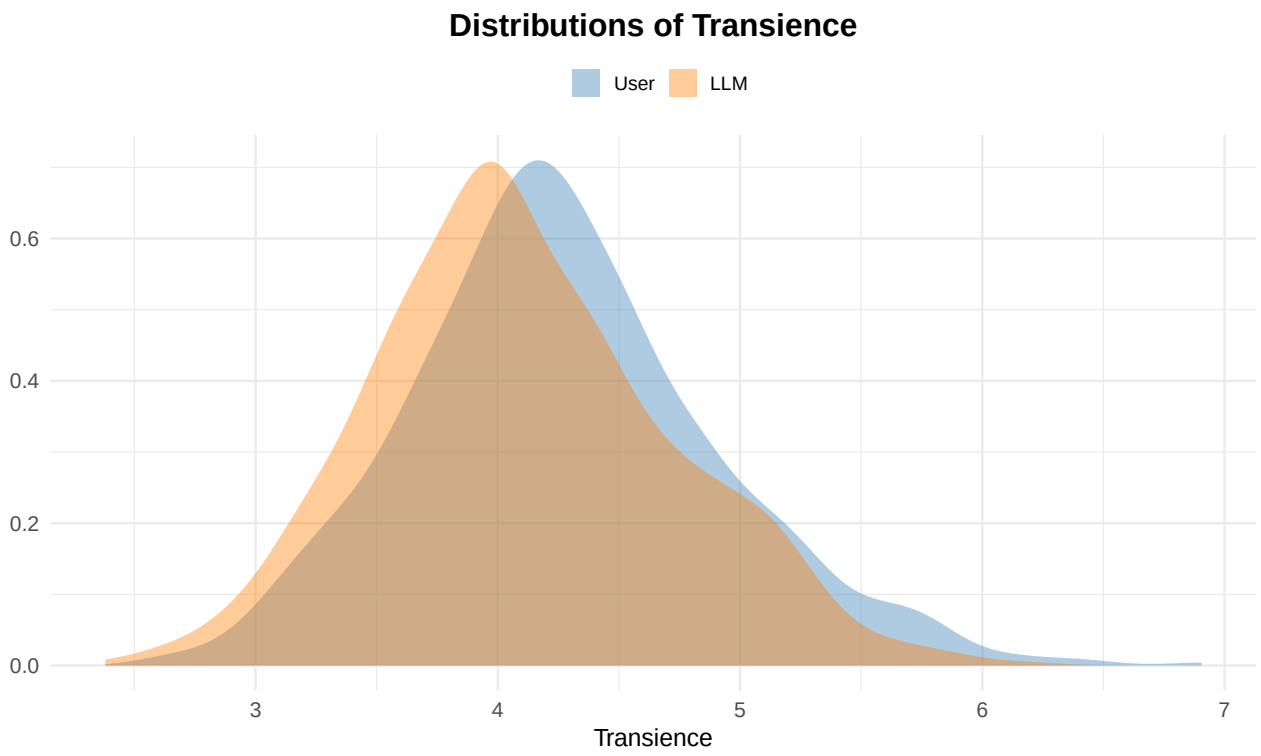
1.1 1.1 Load Data

1.2 1.2 Reshape Data

1.3 1.3 Descriptive Statistics

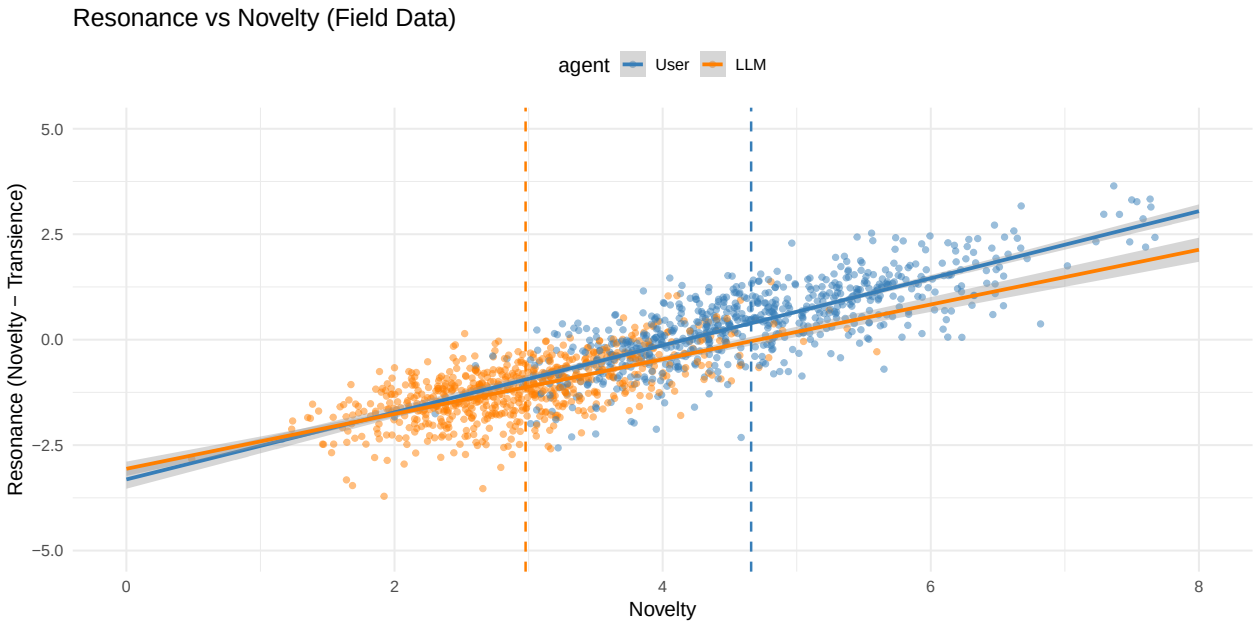
Table 1: Descriptive Statistics for Novelty Metrics by Agent

agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	695	4.660	0.976	4.269	0.640	0.391	0.986
LLM	695	2.978	0.867	4.099	0.615	-1.121	0.837

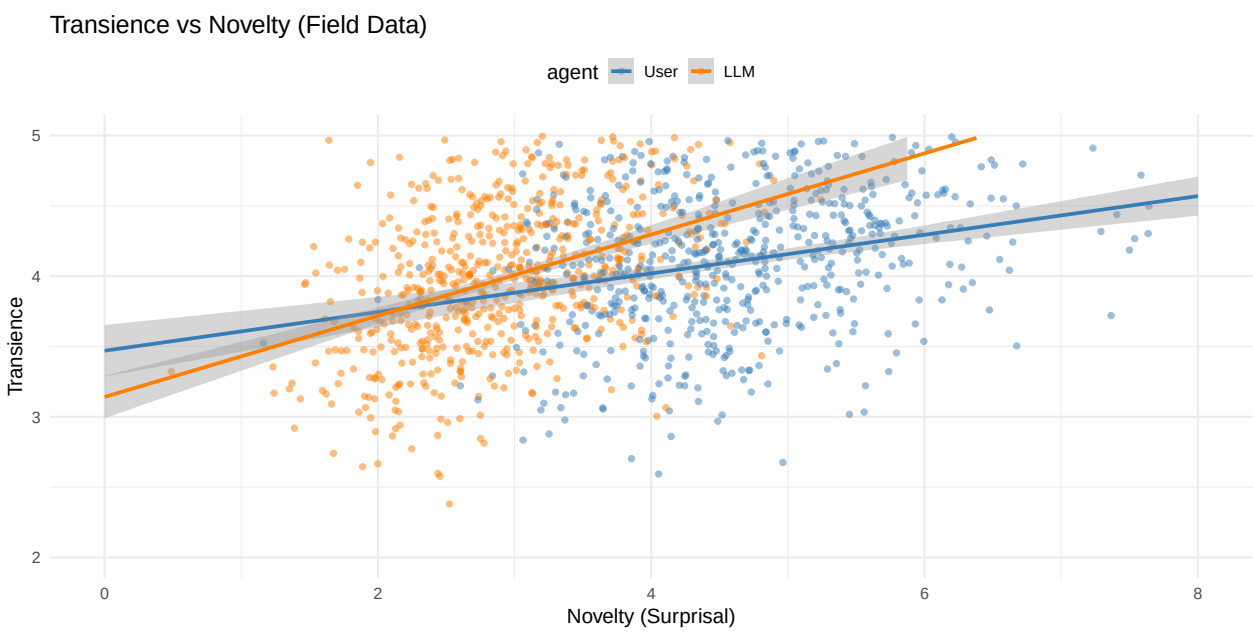


2 2. Novelty and Resonance Analysis

2.1 2.1 Resonance vs Novelty Plot



2.2 2.2 Transience vs Novelty Plot



3 3. Statistical Models

3.1 3.1 Resonance Model

```
## === Resonance Model ===

##                     Estimate Std. Error        df    t value      Pr(>|t|)
## (Intercept)       -4.47774029 0.09415287  527.6204 -47.558191 6.326083e-193
```

```
## surprise           1.04431387 0.01600024 1317.8435  65.268623  0.000000e+00
## agentLLM           0.39702655 0.09235402 1313.5943   4.298963  1.842720e-05
## surprise:agentLLM -0.05094308 0.02373690 1315.0153  -2.146156  3.204309e-02

##
## === Slope Comparison (Resonance) ===

##  agent surprise.trend     SE   df lower.CL upper.CL
##  User           1.044 0.0160 1319    1.013     1.08
##  LLM            0.993 0.0184 1323    0.957     1.03
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3.2 3.2 Transience Model

```
## === Transience Model ===

##                      Estimate Std. Error        df   t value      Pr(>|t|)
## (Intercept)        4.47774029 0.09415287  527.6204 47.558191 6.326271e-193
## surprise          -0.04431387 0.01600024 1317.8435 -2.769574  5.691709e-03
## agentLLM          -0.39702655 0.09235402 1313.5943 -4.298963  1.842720e-05
## surprise:agentLLM  0.05094308 0.02373690 1315.0153  2.146156  3.204309e-02

##
## === Slope Comparison (Transience) ===

##  agent surprise.trend     SE   df lower.CL upper.CL
##  User        -0.04431 0.0160 1319  -0.0757  -0.0129
##  LLM          0.00663 0.0184 1323  -0.0294   0.0427
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4 4. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Copenhagen
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.0   here_1.0.2      afex_1.5-1      cowplot_1.2.0
##  [5] patchwork_1.3.2 zoo_1.8-15      stringi_1.8.7   arrow_22.0.0.1
```

```
##  [9] lmerTest_3.1-3  lme4_1.1-38     Matrix_1.7-1    lubridate_1.9.4
## [13] forcats_1.0.1   stringr_1.6.0   dplyr_1.1.4     purrr_1.2.0
## [17] readr_2.1.6     tidyr_1.3.1     tibble_3.3.0    ggplot2_4.0.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        S7_0.2.1
##  [4] fastmap_1.2.0       TH.data_1.1-2       pacman_0.5.1
##  [7] digest_0.6.39       timechange_0.3.0    estimability_1.5.1
## [10] lifecycle_1.0.4     survival_3.7-0      magrittr_2.0.4
## [13] compiler_4.4.2      rlang_1.1.6         tools_4.4.2
## [16] yaml_2.3.12         knitr_1.51          labeling_0.4.3
## [19] bit_4.6.0           plyr_1.8.9          RColorBrewer_1.1-3
## [22] abind_1.4-8         multcomp_1.4-26     withr_3.0.2
## [25] numDeriv_2016.8-1.1 grid_4.4.2          xtable_1.8-4
## [28] scales_1.4.0        MASS_7.3-61         tinytex_0.58
## [31] cli_3.6.5           mvtnorm_1.3-3       rmarkdown_2.30
## [34] crayon_1.5.3        ragg_1.5.0          reformulas_0.4.2
## [37] generics_0.1.4      rstudioapi_0.17.1   reshape2_1.4.5
## [40] tzdb_0.5.0          minqa_1.2.8         splines_4.4.2
## [43] assertthat_0.2.1    parallel_4.4.2      vctrs_0.6.5
## [46] boot_1.3-31         sandwich_3.1-1      carData_3.0-5
## [49] car_3.1-3           hms_1.1.4           pbkrtest_0.5.5
## [52] bit64_4.6.0-1       Formula_1.2-5       systemfonts_1.3.1
## [55] glue_1.8.0          nloptr_2.2.1        codetools_0.2-20
## [58] gtable_0.3.6        pillar_1.11.1       htmltools_0.5.9
## [61] R6_2.6.1            textshaping_1.0.4   Rdpack_2.6.4
## [64] rprojroot_2.1.1     vroom_1.6.7         evaluate_1.0.5
## [67] lattice_0.22-6      backports_1.5.0     rbibutils_2.4
## [70] broom_1.0.11        Rcpp_1.1.0          coda_0.19-4.1
## [73] nlme_3.1-166        mgcv_1.9-1          xfun_0.55
## [76] pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.